

Distribute AI

Distributed AI (DAI) is an approach where multiple intelligent agents or computers collaborate by sharing computation, data or decisions to solve problems more efficiently than a single system.

Evolution of DAI:

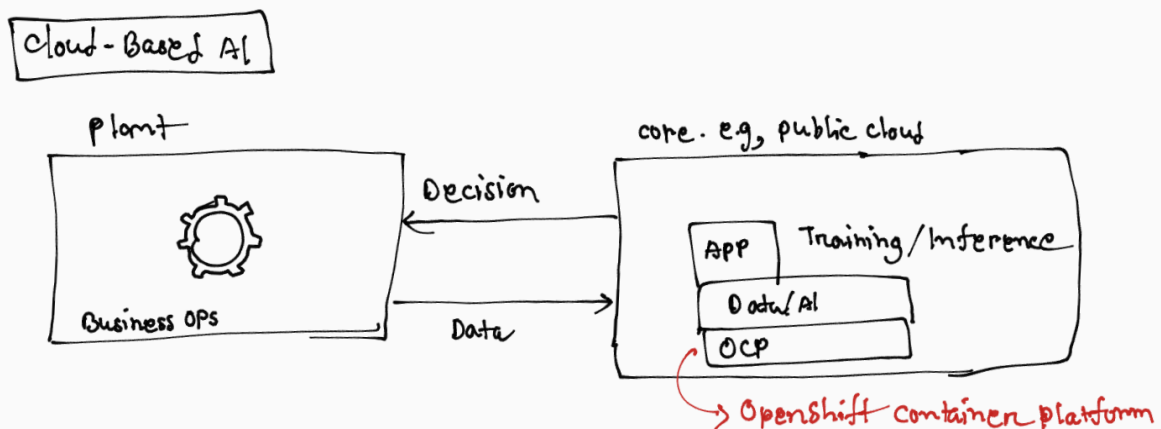
Traditional Centralized AI -> Cloud Based AI -> Edge AI -> DAI

Traditional Centralized AI:

$$AI = f(D)$$

- single computational entity processes Dataset
- Single point of failure.
- Latency issue for geographically distributed users.

Cloud AI:



The cloud infrastructure contains:

1. Application layer
2. Data and AI layer
3. Openshift Container Platform (OCP)

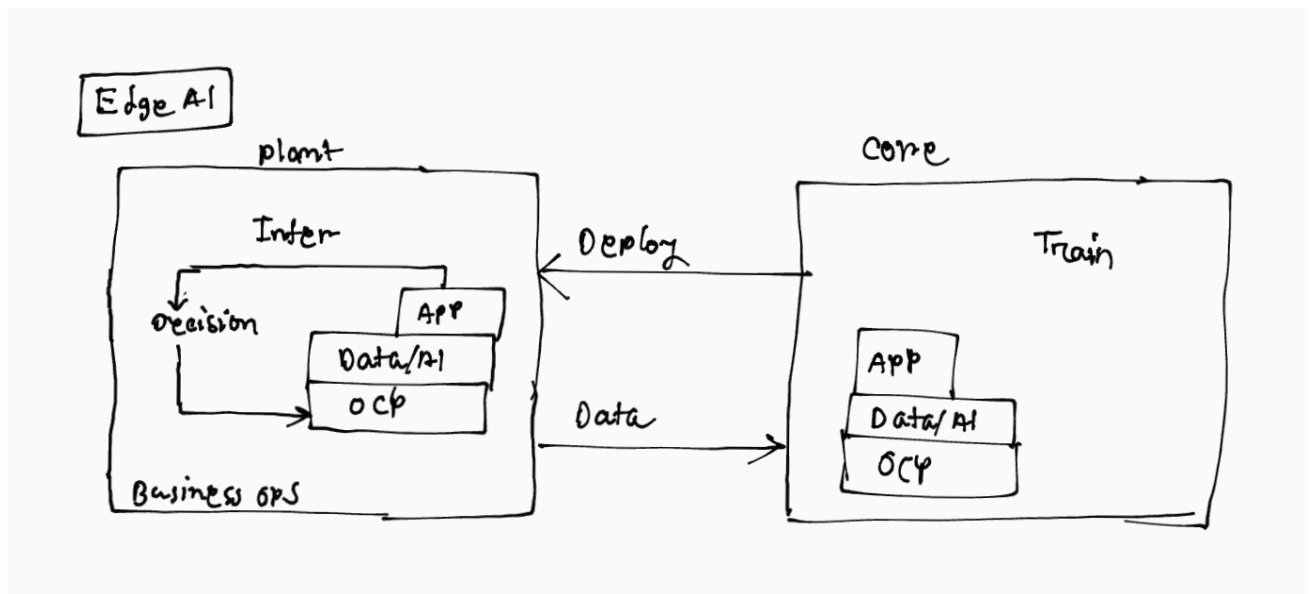
Cloud performs:

1. Model training
2. Model inference
3. Data storage
4. Decision generation

Key Points:

1. Decisions are generated centrally
2. Data is transmitted to the cloud and decisions are sent back to the plant

Edge AI:



Edge AI shifts inference and decision making from the cloud to the edge device or operational environment. That means, localized decision making.

Here,

1. The cloud remains responsible for model training
2. Trained models are deployed to edge locations
3. Inference occurs locally
4. And finally some selected data are sent back to the cloud

So,

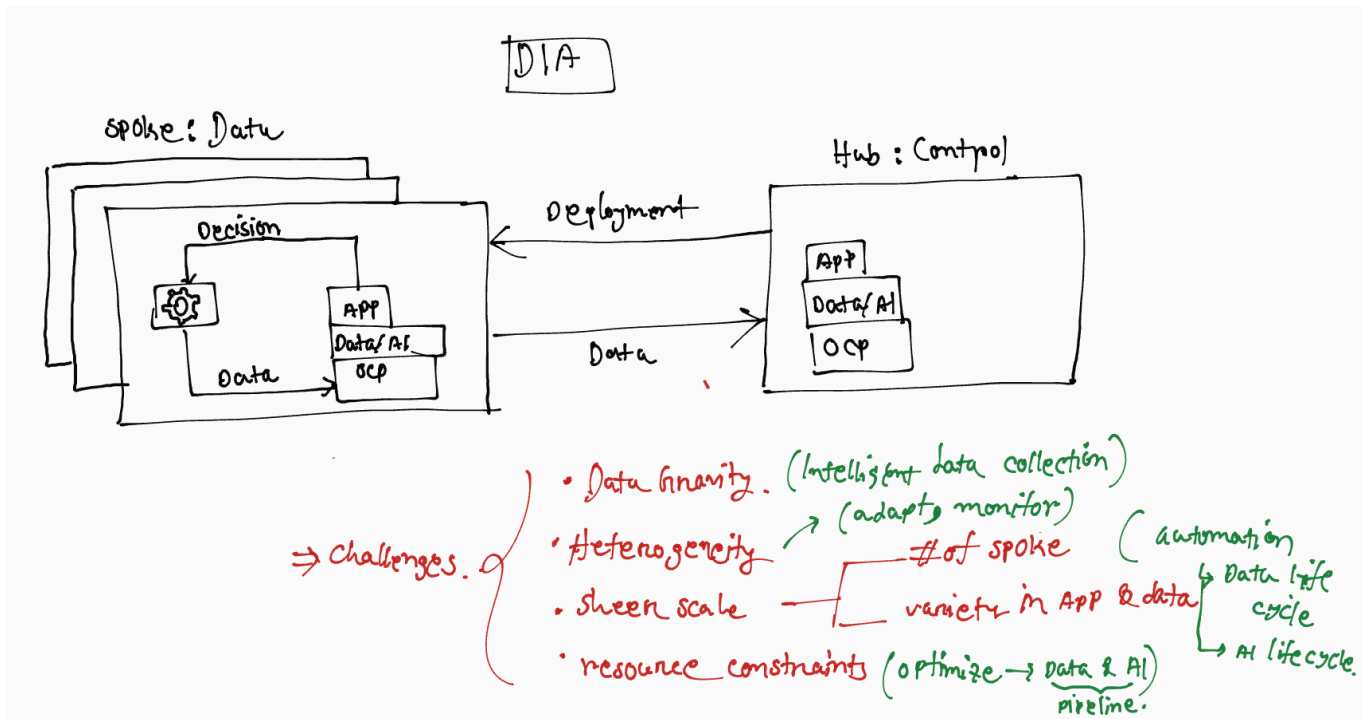
1. Privacy improved
2. Reduced bandwidth usage and
3. Lower latency

Distributed AI:

Most important : where is data and where does it need to be analyzed

Spoke : Where is the data located

Hub : Controls, e.g., Control ai lifecycle from the hub



Responsibilities of **Spoke**:

1. Local data collection
2. Local inference
3. Local decision making
4. Continuous monitoring (helps in addressing Heterogeneity)

Responsibilities of **Hub**:

1. Global coordination
2. Model lifecycle management
3. Training and optimization
4. Deployment management

Key features of DAI:

1. **Distributed intelligence** : Coordinating multiple intelligent nodes DAI ensures Distributed Intelligence
2. **Intelligent data collection**: Adaptive data collection which improves data quality while reducing communication cost.
3. **Heterogeneity**: DAI environment contains diverse components. So DAI must operate across heterogeneous environments
4. **Scalability**
5. **Resource** awareness.